Rigid / Euclidean – Preserves distances + angles
Similitudes – " angles
Linear T. – $\alpha(p+q) = \alpha(p)+\alpha(q)$
$\alpha(ap) = a\alpha(p)$

Affine – preserves parallel lines
Projective " lines

$$x' = ax+by+c$$
$$y' = ex+fy+d$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ e & f \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix}$$ } Cartesian formulation

$$p' = Mp + t$$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & c \\ e & f & d \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$ } Homogenous formulation

$$p' = Mp$$

Translation : $P' = P+T$
Scaling : $P' = S \cdot P$
Rotate : $P' = R \cdot P$
} Non Homogenous

Composition is difficult to expressfon. since translation not expressed as a matrix multiplication.

w/ Homogenous all 3 can be expressed w/ MM by 3×3 matrices
$w = 1 \rightarrow$ affine transformations

2D = 3×3
3D = 4×4

$$\begin{pmatrix} x' \\ y' \\ z' \\ w' \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$$

Multiplying Homogenous coordinates by an affine matrix keeps w unchanged.

$$\Rightarrow \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Homogenous Visualization

$(0,0,1) \rightarrow (0,0,2)$
$(7,1,1) \rightarrow (14,2,2)$
$(4,5,1) \rightarrow (8,10,2)$
} $\omega = 2$

Translation – 2D

$(4,5)$ & $(7,5) \xrightarrow{T} (7,1)$ & $(10,1)$

T → Translation $(3,-4)$

$x' = x + dx$
$y' = y + dy$

$P = \begin{pmatrix} x \\ y \end{pmatrix}$ $T = \begin{pmatrix} dx \\ dy \end{pmatrix}$

$P' = \begin{pmatrix} x' \\ y' \end{pmatrix}$ $P' = P + T$

Homogenous form

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

Translation – 3D

$x' = x + dx$
$y' = y + dy$
$z' = z + dz$

$$\begin{matrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{matrix} * \begin{matrix} x \\ y \\ z \\ 1 \end{matrix} = \begin{matrix} x' \\ y' \\ z' \\ 1 \end{matrix}$$

$T(dx, dy, dz) * P = P'$

Scaling – 2D

$(4,5)$ & $(7,5) \xrightarrow{S} (2, 5/4)$ & $(7/2, 5/4)$

S → Scaling $(1/2, 1/4)$

$x' = Sx * x$
$y' = Sy * y$

$S * P = P'$

$$\begin{pmatrix} Sx & 0 \\ 0 & Sy \end{pmatrix} * \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x * Sx \\ y * Sy \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

Translations + Rotations

TR ≠ RT

R first & then T.

$P' = TRP$

$= MP$

$= Rp + T$

$$M = \begin{matrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{matrix} * \begin{matrix} R_{11} & R_{12} & R_{13} & 0 \\ R_{21} & R_{22} & R_{23} & 0 \\ R_{31} & R_{32} & R_{33} & 0 \\ 0 & 0 & 0 & 1 \end{matrix}$$

$$= \left(\begin{array}{ccc|c} R_{11} & R_{12} & R_{13} & T_x \\ R_{21} & R_{22} & R_{23} & T_y \\ R_{31} & R_{32} & R_{33} & T_z \\ \hline 0 & 0 & 0 & 1 \end{array}\right) = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix}$$

$P' = RTP$

$= MP = R(P+T) = RP + RT$

Not commutative $TS \neq ST$

But associative $((CB)A)P = (C(BA))P = \underbrace{C(B(AP))}_{\text{one point}} = q$

many points $M = CBA$

$q = MP$

Rotation about $P(h,k)$ of θ : $R_{\theta,P}$

(I) Translate $P(h,k)$ to 0.

(II) Rotate θ wrt 0.

(III) Translate (0,0) to $P(h,k)$

$R_{\theta,P} = T(h,k) * R_\theta * T(-h,-k)$

(I) Translate (0,b) to origin

(II) Rotate $-\theta$

(III) Mirror reflect about x-axis

(IV) Rotate θ

(V) Translate origin to (0,b)

$M_L = T(0,b) * R(\theta) * M_x * R(-\theta) * T(0,-b)$

Shearing

x-axis

$$SHx = \begin{matrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

y-axis

$$SHy = \begin{matrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

xy-axis

$$SH_{xy} = \begin{matrix} 1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

z-axis

~~SHxy (x,y)~~ $SHxy(shx, shy) * P = P'$

$$\begin{pmatrix} 1 & 0 & shx & 0 \\ 0 & 1 & shy & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x + z * shx \\ y + z * shy \\ z \\ 1 \end{pmatrix}$$

Inverse Transformation –

A^{-1} will undo anything A does.

Translation → $T^{-1}_{(dx,dy)} = T_{(-dx,-dy)}$

Rotation → $R^{-1}_{(\theta)} = R_{(-\theta)} = R^T_{(\theta)}$ {flips a matrix over its diagonal}

Scaling → $S^{-1}_{(sx,sy)} = S_{(1/sx, 1/sy)}$

Mirror Reflection → $M^{-1}x = Mx$

$M^{-1}y = My$

Composite Transformations –

$TSP \neq STP$

$(1,1) \xrightarrow{S} (2,2) \xrightarrow{T} (5,3)'$

S → scaling (2,2)

T → Translate (3,1)

M.M. = $T(SP) = P'$

$\downarrow$

$TSP = P'$

$$\begin{matrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{matrix} * \begin{matrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{matrix} = \begin{matrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{matrix} * \begin{matrix} x \\ y \\ 1 \end{matrix} = \begin{matrix} x' \\ y' \\ 1 \end{matrix}$$

Rotation-3D

* angle of R
* axis of R

$R_{\theta,k} * P = P'$

ABOUT z AXIS-

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \\ z \\ 1 \end{pmatrix}$$

ABOUT y AXIS-

$$\begin{pmatrix} \cos\theta & 0 & +\sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x\cos\theta + z\sin\theta \\ y \\ -x\sin\theta + z\cos\theta \\ 1 \end{pmatrix}$$

ABOUT x AXIS-

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y\cos\theta - z\sin\theta \\ y\sin\theta + z\cos\theta \\ 1 \end{pmatrix}$$

Mirror Reflection-

$(1,1) \rightarrow (1,-1)$ Reflection about X-axis $\quad x' = x \;\; y' = -y$

$(1,1) \rightarrow (-1,1)$ " " Y-axis $\quad x' = -x \;\; y' = y$

$$M_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad M_y = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

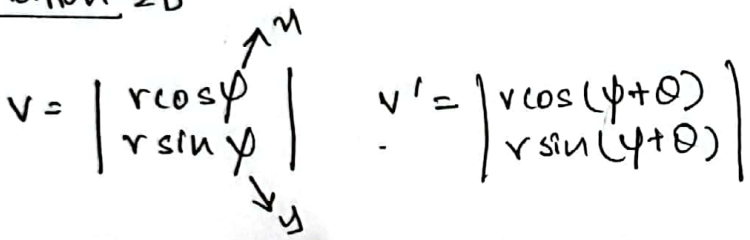
Scaling – 3D

$$x' = s_x * x$$
$$y' = s_y * y$$
$$z' = s_z * z$$

$$S(s_x, s_y, s_z) * P = P'$$

$$\begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix}$$

Rotation – 2D



$$v = \begin{vmatrix} r\cos\psi \\ r\sin\psi \end{vmatrix} \qquad v' = \begin{vmatrix} r\cos(\psi+\theta) \\ r\sin(\psi+\theta) \end{vmatrix}$$

Expand $(\psi+\theta)$ →

$$x' = r\cos\psi\cos\theta - r\sin\psi\sin\theta$$
$$y' = r\cos\psi\sin\theta + r\sin\psi\cos\theta$$

$$\Rightarrow x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$

$$R * P = P'$$

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} * \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\Rightarrow \quad u \quad u \quad u = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

CSE423 – Computer Graphics – AJA

Color Model

Achromatic + Colored Light

Achromatic / Monochrome → B/W

* different intensities of greys
* lacks hue + saturation + brightness
* measured using the quantity of light
* all RGB values are equal
  (R=G=B)
* different values for these 3 generate different shades of greys.

Visible Spectrum – 400nm to 700nm (approx)
Blue – 380nm
Red – 780nm

A color model is a specification of a coordinate system & a subspace within that system where each color is represented by a single point.

RGB (Red, Green, Blue)
CMY (Cyan, Magenta, Yellow)
HSV (Hue, Saturation, Value)
HLS (Hue, Lightness, ~~Value~~ Saturation)

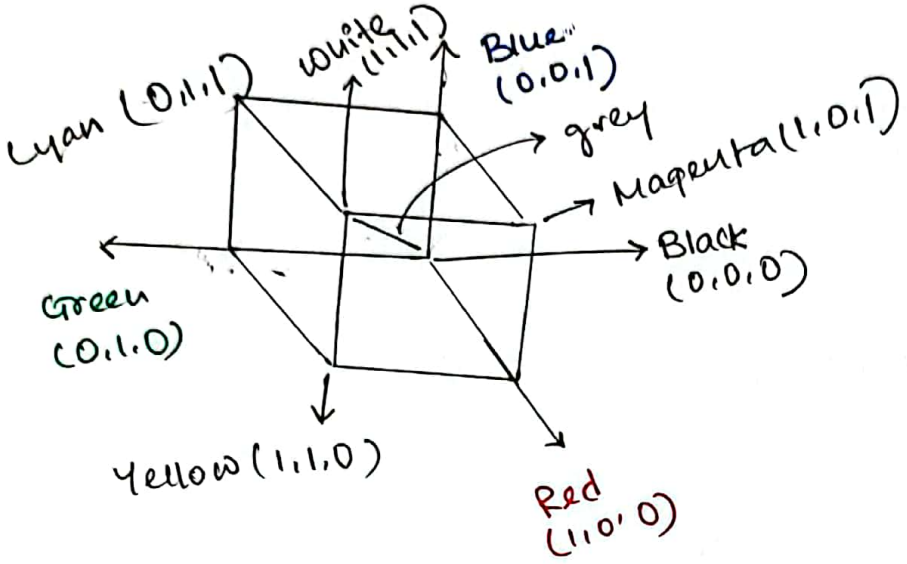
Painting – mixed colors are achieved by subtracting color method.
For computers – additive color method
ie ^

Colors on screen are created w/ light using the additive color method. Begins w/ black & ends w/ white. As more colors are added, the result is lighter & tends to white.

Painting – opposite : Subtractive Color Method
Begins w/ white & ends w/ black.
↑ more colors, tends to black.

RGB Model (TVs + Computers)

* Based on Cartesian Coordinates System



Here, black → absence of any color
white → presence of all colors

* values [0,1]
* represented by a unit cube w/ axes R G B & one corner at the origin of a 3D color coordinate system (Black)
* Grey scale → equal R, G & B.

~~CYK Model~~

CMY Model (Cyan, Magenta, Yellow) ← secondary wrt RGB

↳ primary in subtractive model † RGB → secondary

$C = Y = M \rightarrow$ Black
Muddy Black when printed

↓
adding causes colors to not be reflected + thus not be seen.

white = absence
black = presence

* Printers
* Copiers

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1.00 \\ 1.00 \\ 1.00 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

True Black, 4th color black is added = CMYK Model

C	M	Y	Color
0	0	0	White
1	0	0	Cyan
0	1	0	Magenta
0	0	1	Yellow
1	1	0	Blue
1	0	1	Green
0	1	1	Red
0	1	1	Black

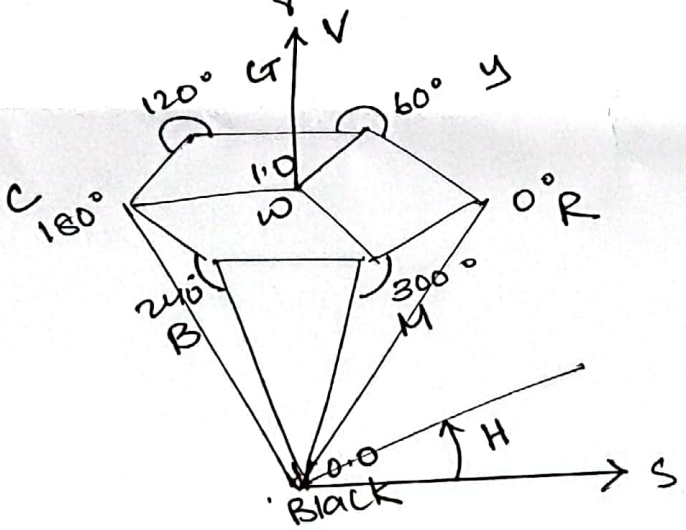
HSV Model

H(Hue) — λ of pure color observed in the signal
↓ dominant color seen
- distinguishes R/Y/G etc
- > 400 hues → human eye

S(degree of dilution) — Inverse of the quantity of "white" present in the signal. Pure color — 100% satn
White + Gray — 0% satn
↓ %
- how far the color is from equal intensity grey
- distinguished R from P etc.
- ≒ 20 satn levels are visible per hue.
- Range [0,1]

V(Value) — Distinguishes grey levels.

* Vision + Image Processing



RGB to HSV Conversion —

Max = max (RGB)
Min = min (RGB)

V = max → Value of HSV → V

if max ≠ 0
    S = max-min/max S → Saturation of HSV
else
    S = 0

if S = 0
    H = undefined H → Hue of HSV
else

    Δ = max-min
    if r = max
        H = g-b/Δ
    elif g = max
        H = 2+(b-r)/Δ
    elif b = max
        H = 4+(r-g)/Δ

    H = H*60
    if H < 0
        H = H+360

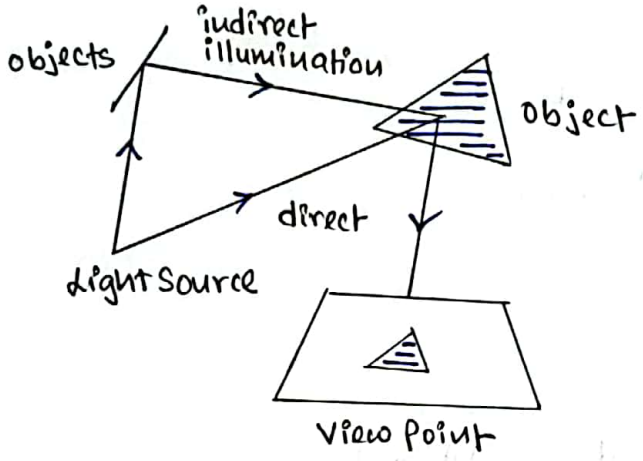
R	G	B	Hue	Saturation	Value	C	M	Y
0.25	0.30	1.00	236°	75%	100%	0.75	0.70	0.0
0.01	1.00	0.99	179.4°	99%	100%	0.99	0.00	0.01
1.00	0.11	0.01	61°	99%	10-0%	0.00	0.89	0.99

CSE423 – Computer Graphics – AJA

Lecture / Chapter 06 – Lighting Model

Illumination – transport of energy from light sources to surfaces + points
- Local
- Global

Illumination Model – Express the factors determining a surface's color / luminous intensity at a particular 3D point.
(outgoing / reflected light)



2 components of illumination –
(I) Light sources
(II) surface properties

* Light sources are described by a luminance / intensity "I"
  - each color is described separately
  - $I = [I_R \; I_G \; I_B]$

* Types of light sources –
  (I) Ambient
  (II) Diffuse
  (III) Spot

Dark object → Ambient Reflection (colour) + Diffuse Reflection (some defn) = Correct color + object ← object + Specular Reflection (shininess)

Ambient light
- no identifiable source or direction
- product of multiple reflections of light from many surfaces present in the environment
- computationally inexpensive

Ambient light
- Global
  - independent of L's.
  - Lights entire scene
  - Ex. reflection of sunlight from several surfaces
- Local
  - from added light sources
  - can be different for each light + primary color
  - Ex. reflection of fluorescent lamps from several surfaces

* Ambient Reflection

- reflection of light that does not come directly from a light source
  * In darkness, we can make out the edges of objects (light bounces off of objects)
  * Phong - Direct lighting Model (assume that ambient light falls equally on all objects).

$$A = I_a k_a$$

I_a → intensity of the ambient light
k_a → ambient coefficient [0,1]

- is set to provide right amount of ambient light for a scene.
  - $k \to 1$ Bright scenes
  - $k \to 0$ Dark scenes

* Ambient Reflection Coefficient

  * Effect of adding ambient light to the diffuse light reflected by a sphere

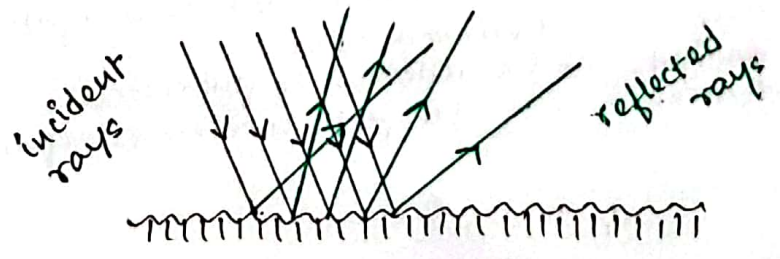
- Diffuse source intensity → 1.0
- " reflection coefficient → 0.4
- Ambient source intensity → 1.0
- " reflection coefficient → 0.0 0.1 0.3 0.5 0.7

↓ ambient light ; shadows too deep + harsh
↑ " " ; picture washed out + bland

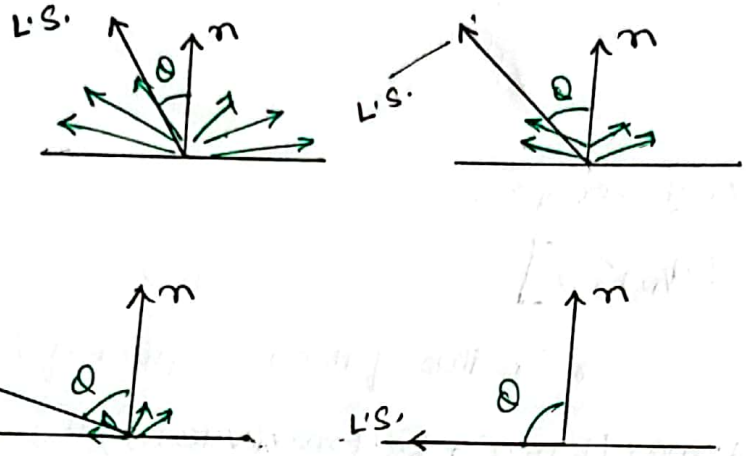


Diffuse Reflection -

* || rays from L.S.
* Reflected rays are scattered (not smooth surface)
* direction of reflection
* some are visible, some are not



Diffuse Reflection – Phong's Model

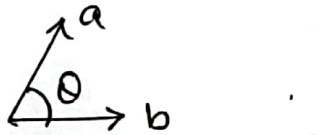


* Light reflected equally in all direction
* magnitude of reflection depends on θ

Phong's Diffuse Reflection —

$$D = I_p k_d \max[\cos(\theta), 0]$$

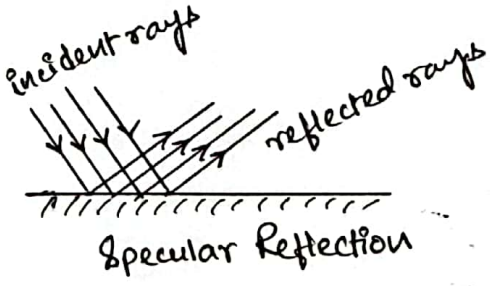
- I_p: intensity of L.S.
- k_d: $[0,1]$ diffuse coefficient
- $\max[\cos(\theta), 0]$: no light is reflected if L.S. is behind the surface

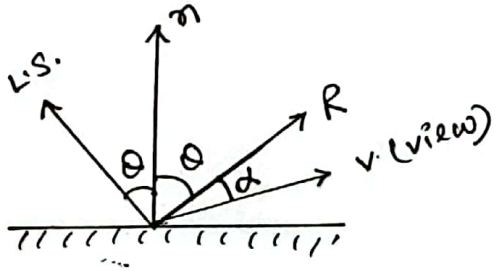


$a \cdot b = |a| \cdot |b| \cos\theta$

$\therefore \cos\theta = L \cdot n$ (unit vectors)

$$D = I_p k_d \max[L \cdot n, 0] \quad — (1)$$





Phong's Specular Model

amount of light being reflection

$$S = I_p k_s \cdot \cos^n(\alpha) \quad — (2)$$

$k_s \rightarrow [0,1]$ specular coefficient

$n \rightarrow$ specular exponent/shininess

$\alpha \rightarrow$ angle betn R & V

$\therefore \cos\alpha = V \cdot R$ (unit vectors)

where $\bar{R} = 2(\hat{L} \cdot \hat{n})\hat{n} - \hat{L} \quad — (3)$

$$I = I_a k_a + I_p k_d \max(L \cdot n, 0) + I_p k_s (\max(V \cdot R, 0))^n$$

- $I_a k_a$ → Ambient light
- $I_p k_d \max(L \cdot n, 0)$ → Diffuse light
- $I_p k_s (\max(V \cdot R, 0))^n$ → Specular light

Attenuation – loss of light energy over space

$$f_{att} = \max\left[1 - (d/r)^2, 0\right]$$

r = radius of the L.S. sphere of influence

$$I = I_a k_a + I_p f_{att} (k_d \cdot \max(L \cdot n, 0) + k_s \cdot (\max(V \cdot R, 0))^n)$$

For multiple L.S.,

$$I = I_a k_a + \sum_{i=1}^{m} I_{pi} f_{att} (k_d \cdot \max(L_i \cdot n, 0) + k_s (\max(V \cdot R_i, 0))^n)$$

— x —

Ex. A light source w/ intensity 5 and radius of influence 50 is located at (2,3,4) from which you are called to calculate the illumination of a point on the xy plane. The camera is set at a point (5,6,3) & the light is reflected back from (4,4) of the plane. The ambient, diffuse and specular coefficient is given at 0.2, ~~0.3~~ 0.5 and 0.4 respectively.

(a) Represent the reflected ray R in the unit vector
(b) Specular Reflection Intensity for a shininess factor of 10
(c) Attenuation Factor
(d) If ambient light ~~intensity~~ intensity is at 2, calculate the total reflected light intensity at the given point according to Phong's model w/ the attenuation factor.

(a) $L = (2,3,4) - (4,4,0)$

$= (-2,-1,4)$

$\hat{L} = \frac{-2\hat{i} - \hat{j} + 4\hat{k}}{\sqrt{21}}$

xy plane
normal = z-axis
$= \hat{k}$

$\hat{L} \cdot \hat{n} = \left(\frac{-2\hat{i} - \hat{j} + 4\hat{k}}{\sqrt{21}}\right) \times \hat{k}$

$= \frac{4}{\sqrt{21}}$

$\bar{R} = 2(\hat{L} \cdot \hat{n})\hat{n} - \hat{L}$

$= \frac{8}{\sqrt{21}}\hat{k} - ~~(-2,-1,4)~~ \left(\frac{-2\hat{i} - \hat{j} + 4\hat{k}}{\sqrt{21}}\right)$

$= \frac{1}{\sqrt{21}}(+2\hat{i} + \hat{j} - 4\hat{k} + 8\hat{k})$

$= \frac{1}{\sqrt{21}}(2\hat{i} + \hat{j} + 4\hat{k})$

$\hat{i} \cdot \hat{i} = 1$

(b) $\bar{V} = (5,6,3) - (4,4,0)$

$= (1,2,3)$

$\hat{V} = \frac{1}{\sqrt{14}}(\hat{i} + 2\hat{j} + 3\hat{k})$

$\hat{R} \cdot \hat{V} = \frac{1}{\sqrt{21}} \cdot \frac{1}{\sqrt{14}}(2\hat{i} + 2\hat{j} + 12\hat{k})$

$= \frac{1}{\sqrt{294}}(16) = \frac{16}{\sqrt{294}}$

$g = I_p \cdot k_s \cdot \left(\frac{16}{\sqrt{294}}\right)^n$

$= 5 \times 0.4 \times \left(\frac{16}{\sqrt{294}}\right)^{10}$

(c) $d = [(2-4)^2 + (3-9)^2 + (4-0)^2]^{1/2}$

$= \sqrt{21}$

$f_{att} = \max\left(1 - \left(\frac{d}{50}\right)^2, 0\right)$

$= \max\left(1 - \left(\frac{\sqrt{21}}{50}\right)^2, 0\right)$

$= 0.9916$

(d) $I = I_a k_a + I_p f_{att}\left(k_d \max(\hat{L}\cdot\hat{n}, 0) + k_s(\max(\hat{R}\cdot\hat{V}, 0))^n\right)$

$= 2 \times 0.2 + 5 \times 0.9916\left(0.5 \times \frac{4}{\sqrt{21}} + 0.4 \times \left(\frac{16}{\sqrt{294}}\right)^{10}\right)$

$= ~~0.8333~~ \; 3.575$

(Ans)

CSE423 – Computer Graphics – AJA

Lecture 11+12+13 – Projection

Projection – a process by which we can transform points in a coordinate system of dimension "n" (3D) into points in a coordinate system of dimension less than "n" (2D).

3D Model (I/P) → Projection → 2D Model (O/P)

Centre of Projection → c.o.p.

* Projection → onto a plane
* Projectors → straight lines

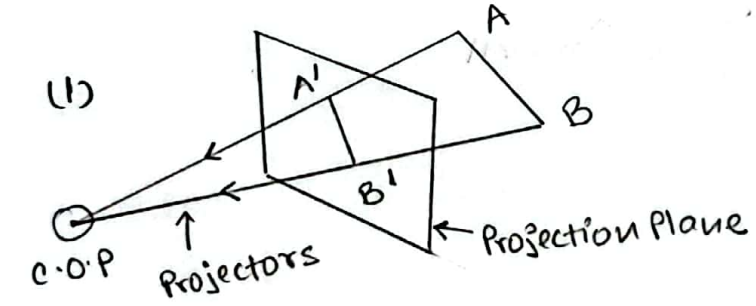
3D to 2D Projection is defined by straight projection rays (projectors) emerging from the c.o.p. passing through each point of the object & intersecting the "projection plane" to form a projection

2 types of Projection –

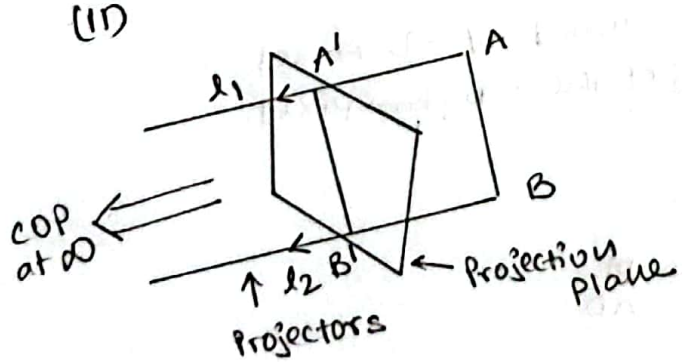
(I) Perspective (distance to c.o.p is finite)

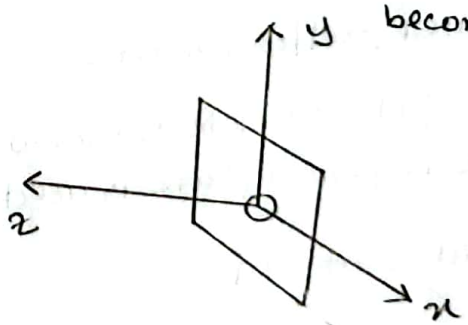
(II) Parallel (∞)
– l_1 & l_2 rays of sight
– inc. d to ∞, the rays become ||.

(I)



(II)

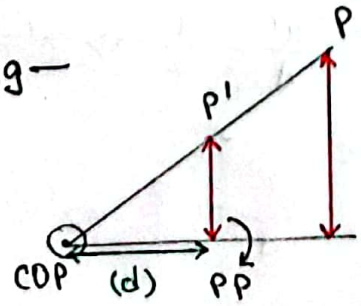




Monitor in 3D-view.

–ve z axis inside & +ve z axis outside the plane.

Simplifying –



d → distance from c.o.p to P.P.

$|P'| = |P_p| \cdot |P|$ → object

$|P_p|$ → projection matrix

$|P'|$ → projection

Based on the orientation of PP:

(I) Orthographic Projection—

z-axis perpendicular to the projection plane



(II) Oblique Projection—

z-axis not perpendicular to the projection plane

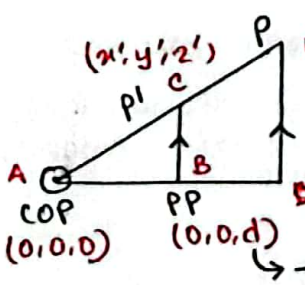
(excluded from the syllabus)

Perspective	Parallel
— human vision	— less realistic due to no foreshortening
— C.O.P. distance ↑ object size ↓ (inversely proportional)	— ∥ lines remain ∥
— angles remain intact for faces ∥ to projection plane.	— ↗ as well
— ∥ lines do not project ∥ lines	

P' → final pixel/coordinate

for an object, calculations + view are determined w/ 3D Model.
And the projection plane is used to project the 2D plane pixel.

⊛ Origin is at COP.



Similar Δs,

$$\frac{BC}{AB} = \frac{DE}{AD}$$

-ve since it goes to the -ve z axis; z is also -ve.

x':

$$\frac{x'}{d} = \frac{x}{z} \Rightarrow x' = x/(z/d)$$

z':

$$\frac{z'}{d} = \frac{z}{z} \Rightarrow z' = z/(z/d)$$

y':

$$\frac{BC}{AB} = \frac{DE}{AD} \Rightarrow \frac{y'}{d} = \frac{y}{z}$$

$$\Rightarrow y' = y/(z/d)$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/p & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$1/p \rightarrow ?$

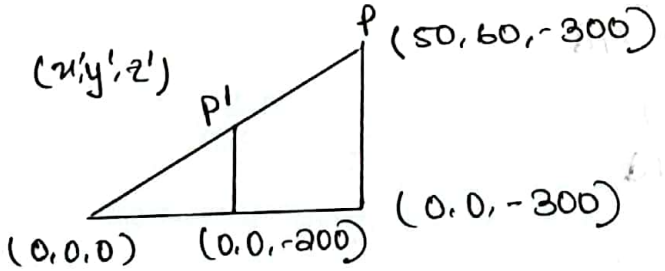
projection matrix

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ z/d \end{bmatrix} \rightarrow \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x'/z/d \\ y'/z/d \\ z'/z/d \\ 1 \end{bmatrix}$$

$\hookrightarrow x \hookleftarrow$

$\Rightarrow$ To make it match, we need all terms in z/d (divide)

Ex. origin at COP, calculate projected point coordinates for a point (50, 60, -300) where the projection plane is at a distance of 200 from the COP?

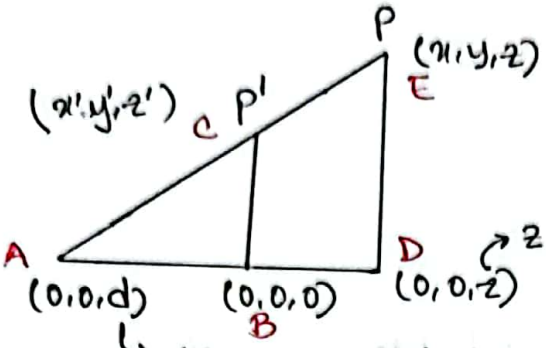


$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/200 & 0 \end{bmatrix} \begin{bmatrix} 50 \\ 60 \\ -300 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 50 \\ 60 \\ -300 \\ 3/2 \end{bmatrix} \rightarrow \text{Divide by } 3/2$$

$$= \begin{bmatrix} 33.33 \\ 40 \\ -200 \\ 1 \end{bmatrix} \quad \therefore P' = (3333, 40, -200)$$

④ Origin is at PP.



z → -ve as it is in the -ve z-axis's

↳ d is +ve as it is towards the +ve z-axis

$$y' \rightarrow \frac{BC}{AB} = \frac{DE}{AD}$$

$$\Rightarrow \frac{y'}{d} = \frac{y}{d-z} \rightarrow z = -ve \text{ so actually an addn.}$$

$$\Rightarrow y' = \frac{y}{d-z/d} = \frac{y}{1-z/d}$$

$$x' \Rightarrow x' = \frac{x}{d-z/d} = \frac{x}{1-z/d}$$

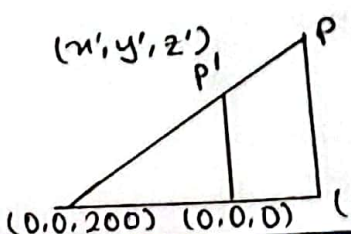
$$z' \Rightarrow z' = 0 \text{ (origin at PP)}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/d & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Projection Matrix

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \\ 1-z/d \end{bmatrix} \Rightarrow \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x/1-z/d \\ y/1-z/d \\ 0 \\ 1 \end{bmatrix}$$

Ex. for origin at PP, calculate projected point coordinates for a point (50, 60, -100) where d = 200 from COP?



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/200 & 1 \end{bmatrix} \cdot \begin{bmatrix} 50 \\ 60 \\ -100 \\ 1 \end{bmatrix} = \begin{bmatrix} 50 \\ 60 \\ 0 \\ 1.5 \end{bmatrix}$$

$$\Rightarrow \begin{matrix} x' \\ y' \\ z' \\ 1 \end{matrix} = \begin{matrix} 50/1.5 \\ 60/1.5 \\ 0 \\ 1 \end{matrix} = \begin{matrix} 33.33 \\ 40 \\ 0 \\ 1 \end{matrix}$$

$\therefore P'(33.33, 40, 0)$

↳ one plane is fixed w/ a fixed value

Ⓐ Orthographic Projection Matrix —

$$P_m = \begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{matrix}$$

} Projection on xy plane while $z = 0$.

$$P_m * \begin{matrix} 2 \\ 4 \\ 9 \\ 1 \end{matrix} = \begin{matrix} 2 \\ 4 \\ 0 \\ 1 \end{matrix} \rightarrow (2,4) \quad [3D \text{ to } 2D]$$

Ⓐ Sometimes z is fixed to a certain value.

$z = 1$

$$\begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{matrix} \rightarrow P_m * \begin{matrix} 2 \\ 4 \\ 9 \\ 1 \end{matrix} = \begin{matrix} 2 \\ 4 \\ 0+1 \\ 1 \end{matrix} = \begin{matrix} 2 \\ 4 \\ 1 \\ 1 \end{matrix}$$

Ⓐ Suppose, yz plane & x is fixed at $x = 6$.

$$\begin{matrix} 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix} * \begin{matrix} x \\ y \\ z \\ w=1 \end{matrix} = \begin{matrix} 6 \\ y \\ z \\ 1 \end{matrix}$$

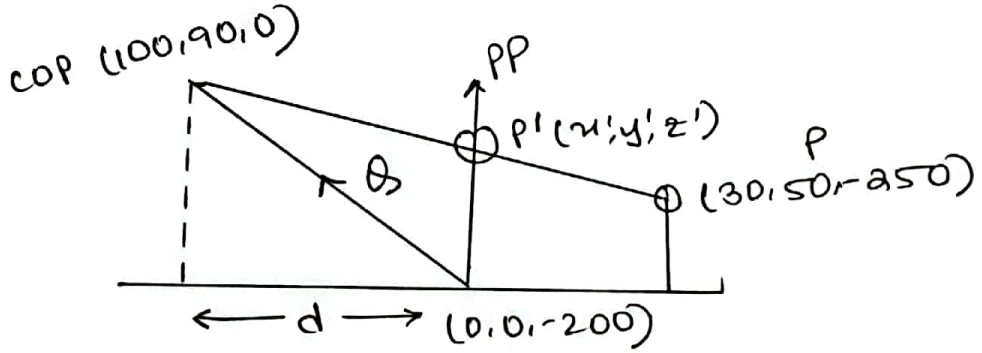
Given plane — consider their values ; ignore the others

Ex. xz plane → find for x & z ; ignore y.

General Perspective Matrix —

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\frac{Q_{dx}}{Q_{dz}} & z_p \cdot \frac{Q_{dx}}{Q_{dz}} \\ 0 & 1 & -\frac{Q_{dy}}{Q_{dz}} & z_p \cdot \frac{Q_{dy}}{Q_{dz}} \\ 0 & 0 & -\frac{z_p}{Q_{dz}} & z_p \cdot \left(1 + \frac{z_p}{Q_{dz}}\right) \\ 0 & 0 & -1/Q_{dz} & 1 + z_p/Q_{dz} \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Calculate the projected point coordinates for a point (30, 50, -250) where COP is at (100, 90, 0) & d = 200 from the COP.



$Q_{dx} = 100$

$Q_{dy} = 90$

$Q_{dz} = 200$

$z_p = -200$

$\therefore P' = (44, 58, -200)$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1/2 & -100 \\ 0 & 1 & -9/20 & -90 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/200 & 0 \end{bmatrix} * \begin{bmatrix} 30 \\ 50 \\ -250 \\ 1 \end{bmatrix}$$

$$= \begin{matrix} 30+125-100 \\ 50+112.5-90 \\ -250 \\ 514 \end{matrix} = \begin{matrix} 55 \times 4/5 \\ 72.5 \times 4/5 \\ -250 \times 4/5 \\ 514 \times 4/5 \end{matrix} = \begin{matrix} 44 \\ 58 \\ -200 \\ 1 \end{matrix}$$